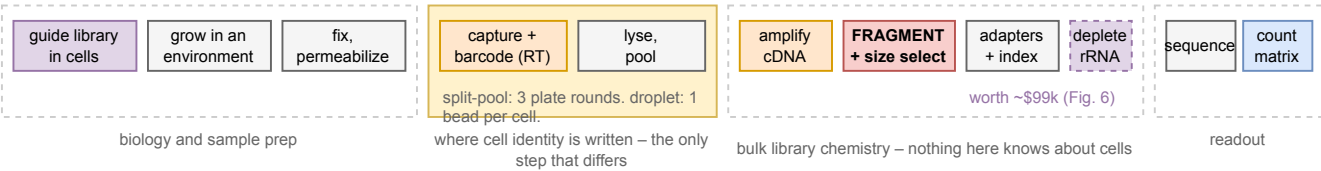
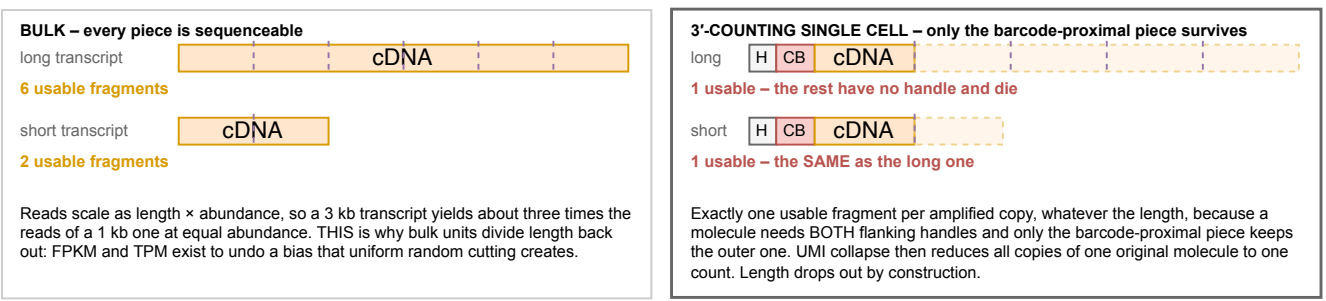
The whole experiment, and where each stage can lose you cells, molecules or identity

Read left to right. The wet-lab stages are the same for both method families; what differs is only how cells are kept apart while their barcodes are written (Figs. 2 to 4). Fragmentation is drawn explicitly because it is the step that decides whether a count means “molecules” or “molecules times length”, and because ribosomal RNA depletion is placed relative to it (Fig. 6).

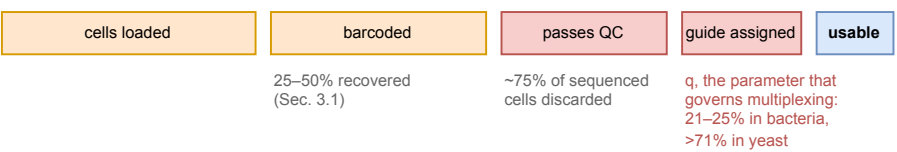
a The stages. Only the shaded block differs between split-pool and droplet.



b Fragmentation in bulk RNA-seq and in a 3′-counting single-cell assay. Same cutting, opposite consequence.



c What each stage costs you. The funnel, not the chemistry, is what a budget is made of.



and separately, what happens to a READ

Of 100 reads bought for an undepleted split-pool run, about 10 are PhiX or an unresolvable barcode, 94 of the remainder are ribosomal, and three quarters of the cells they came from are discarded. Roughly 1 read in 100 survives all four tolls; depletion takes it to about 11.

Segments transcript / cDNA cell barcode (CB) or rRNA enzyme, probe or cut UMI / output H = handle, adapter container faded + dashed = discarded

Panel b is the reason this figure exists. Bulk and single-cell protocols perform the SAME physical cutting, and the consequence inverts because of what happens afterwards: in bulk every piece can be sequenced, so reads scale with length; in a 3′-counting assay only the piece carrying the outer handle can be, so exactly one survives per molecule regardless of length. Applying TPM or FPKM to a 10x or SPLiT-seq count matrix is therefore an error, not a stylistic choice. Composed from the protocols as described in Secs. 2.4, 3.1 and 3.2, not traced from any published figure. Every number drawn by hand here is sourced in the provenance table for this figure.